Repeat-Aware Hybrid Recommendation of TV Programmes on sweet.tv: Personalized PageRank, ALS and Association Rules under a Schedule-Constrained MAP@5 Protocol

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Assignment: Midterm Quiz — TV Program Recommendation (sweet.tv Kaggle competition)
Code: https://github.com/PetrenkoMaksim/sweettv-midterm Kaggle user: underqualified

Abstract — We recommend five TV programmes per user for ISO weeks 31–42 of 2020 from sweet.tv channel-viewing logs of weeks 11–30 and the programme schedule, scored by MAP@5. Views are joined to labelled airings by channel and time overlap, and every method is validated on the same temporal split (history weeks 11–20, target weeks 21–30) with the competition's own label rule; we compare popularity, Personalized PageRank (PPR), implicit ALS and association rules, and combine them in a LightGBM re-ranker. A rank-averaged ensemble of four LightGBM re-rankers reaches MAP@5 0.4319 on the 1,238 test users of the validation fold, against 0.3871 for the best single method (PPR) and 0.2688 for popularity; its Kaggle private score is 0.44826 (public 0.45214). Because the label counts repeatedly watched airings, methods that keep a user's own programmes on top dominate pure collaborative filtering.

Index Terms — recommender systems, implicit feedback, personalized PageRank, matrix factorization, association rules, MAP@5

1. Introduction

Problem. For each of 1,260 sweet.tv users we must output a ranked Top-5 of programme IDs (`tv_show_id`) for weeks 31–42. Inputs are a channel-level viewing log (who watched which channel from when to when, weeks 11–30) and the broadcast schedule (which programme aired on which channel when, weeks 11–42). The ground truth, per the competition Data tab [7], is each user's five programmes with the most airings watched for ≥ 80 % of their duration in the target window; the metric is MAP@5. **Motivation.** A "your programmes this week" row is the cheapest home-screen personalisation an IPTV service has; showing what a user returns to shortens the path to playback. **Example.** User `5024631139882254261` watched cartoons in weeks 11–20 ('Вспыш и чудо-машинки' 143 airings ≥ 80 %, 'Свинка Пеппа' 140); the final hybrid's fold-A Top-5 hits 'Свинка Пеппа', 'Вспыш…' and 'Гуппи и пузырики' (AP@5 0.520, Table 3). **Contributions.** (i) A single leakage-free protocol (join → temporal split → competition label rule → MAP@5) applied identically to five methods and 63 logged runs (`results/attempts.csv`); (ii) evidence that on this label, repeat-preserving graph scores beat matrix factorisation by a wide margin, and an ensemble of re-rankers with schedule-aware (channel-affinity) candidates adds +0.045 MAP@5 on top.

2. Related Work

[1] implicit-feedback ALS with confidence weights c = 1 + α·r (our ALS confidence); [2] Apriori support/confidence and downward closure (our pairwise rules over "watched ≥ 80 %" baskets); [3] PageRank with a personalised teleport vector (our PPR restart); [4] two-stage candidate generation → ranking (our hybrid); [5] LightGBM (our re-rankers); [6] the `implicit` library (our ALS); [7] the competition, data and label rule. *Alternatives considered.* BPR-MF — not used, ALS already shows that pure CF misses the repeat signal; item-kNN — PPR restarted at the user covers the same 2-hop neighbourhood; content-based filtering on genres/titles — dropped for time; category and duration enter the re-ranker instead.

3. Method

Data and matching rule. View log (`user_id, vsetv_id, start_time, stop_time, duraton`) × schedule (`channel_id, tv_show_id, start_time, duration, title, category, genres, …`). A view matches an airing if `vsetv_id = channel_id` and the intervals overlap; overlap seconds are summed per (user, airing) and clipped to the airing duration (24,006 of 3,014,129 pairs were clipped); screen time `frac = overlap / duration`; the week is the ISO week of the airing start. **Missing programme IDs:** rows with `tv_show_id = 0` (unlabelled airtime, 89 % of schedule rows, 83–84 % of airtime) are not items and are dropped before the join; 94.1 % of watched seconds land on some airing, 53.1 % on labelled airtime (`results/join_checks.txt`). `user_id` is a 64-bit hash, read as a string (never float). A timezone check (view starts within 60 s of an airing start, clock shifts −3…+3 h) peaks weakly at 0 h (3.63 % vs ≤ 3.53 %), so no offset is applied. **Programme metadata used:** channel, airing start and duration (join, candidate pool, airings-in-target-window features), category (feature), titles (presentation only); genres, director, actors and year are not used — content-based filtering was not implemented.

	rows	users	channels / shows	note
view log weeks 11–30	5,821,987	4,914	202 channels	3,014,129 (user, labelled airing) pairs

	rows	users	channels / shows	note
schedule file 11-20 (labelled / total)	115,613 / 1,074,235		400 ch. / 5,847 shows	labelled airtime 16.0 %
schedule file 21-30 (labelled / total)	121,505 / 1,074,394		403 ch. / 6,363 shows	labelled airtime 16.9 %
schedule file 31-42 (labelled / total)	144,774 / 1,295,378		404 ch. / 6,636 shows	candidate pool weeks 31–42: 6,642 shows
test users (submission.csv)	1,260			all in the view log; 1,258 with labelled history

Table 1. Data accounting (`results/data_stats.txt`, `results/final_window_stats.txt`). Files overlap at week borders, so all splits use the airing's ISO week.

Validation protocol (identical for all methods). Fold A: history = joined rows of weeks 11–20 → truth = competition rule on weeks 21–30 (count airings with frac ≥ 0.8 per show, top-5 by count, ties by smaller id; `tools/evaluate.py`); candidate pool = labelled shows airing in weeks 21–30 (6,376); MAP@5 with denominator min(5, |truth|) as in Kaggle's ml_metrics. We report (a) all 4,643 users with non-empty truth and (b) the 1,238 test users among them. Users with no history and padding of short lists use popularity (history window, restricted to candidates). Fold B (11–24 → 25–30) checks robustness. Viewing data of the target weeks is used only to build the truth.

Pipeline. `prep.py` join → `protocol.py` split, truth, candidates → `methods.py` scores → `features.py` candidates + features → `ranker.py` LightGBM → `run.py final` → `submissions/` → `tools/check_submission.py`.

Methods. *Popularity* (reference): distinct users with ≥ 1 airing watched ≥ 80 %. *Repeat frequency* (reference): the user's own shows by count of ≥ 80 % airings, ties by seconds. *PPR*: bipartite user–show graph, edge = log(1+seconds) or count of ≥ 80 % airings, row-normalised transitions, restart probability 1−α. Because the walk alternates user → show → user, the show part of the PPR vector is $(1-\alpha)\cdot q\cdot(I-\alpha^2T)^{-1}$ with T = P(show→user)·P(user→show); we solve this exactly (one 7,146 × 7,146 inverse per configuration) instead of iterating, and checked it against 200 power iterations: max abs difference 7.0e-10 (`results/ppr_check.txt`). *ALS* [1], [6]: confidence 1 + a·log(1+sec/60) or 1 + a·count80, 15 iterations, factors {32, 64}, λ {0.01, 0.1}. *Association rules* [2]: baskets = shows with ≥ 1 airing watched ≥ 80 %; frequent shows (support ≥ 5 or 20 users), pairwise rules a→b with confidence ≥ 0.05 or 0.2; a candidate is scored by the max or sum of confidences (or lift) of rules fired by the basket, optionally with the trivial self-rule a→a = 1 so that own shows stay eligible. *Hybrid* [4], [5]: candidates = the user's history shows that air in the target window ∪ top-30 PPR ∪ top-30 ALS ∪ top-30 rules ∪ top-20 popularity ∪ top-50 by channel affinity Σ_c share(u, c)·airings(show on c in the target window) (187.2 per user, recall of truth 0.8428; 0.7909 without the channel source — `results/candidate_recall_*.txt`); 39 features: the user's history with the show (per-week rates of ≥ 80 % airings and seconds; counts in the last 4, 2 and 1 weeks; counts decayed with half-life 1 and 3 weeks; seconds in the last 4 weeks; distinct days; weeks since last watch; share of the show's airings watched; share of the user's ≥ 80 % airings; in the user's top-5 of the last 2 weeks), the schedule (airings per history and per target week, expected = share watched × target airings and its rank within the user, median duration, category, new-show flag), popularity (per week, last 4 weeks, rank), user–channel share and hour/weekday profile overlap, user activity (overall, last 4 weeks, number of shows), and score + within-user rank of PPR, ALS, rules, channel affinity and repeat (rank 9999 for a zero score). Four LightGBM models are rank-averaged within each user: binary (label = show in the fold-A truth), lambdarank on the same label, lambdarank with graded labels (1–5 by log2 of the truth item's ≥ 80 % airings) and Poisson regression on the count of ≥ 80 %-watched airings in the target window, the quantity the label rule ranks by; 3-fold GroupKFold by user so no user is scored by a model that saw it.

Tools, configuration (`results/versions.txt`): Python 3.11.5, numpy 1.26.4, pandas 2.1.1, pyarrow 15.0.1, scipy 1.11.3, scikit-learn 1.2.2, lightgbm 4.7.0, implicit 0.7.3, threadpoolctl 3.6.0; macOS 14.5, Apple M1 Pro, 16 GB RAM. Seed 42 (ALS, LightGBM); LightGBM 400 trees, learning rate 0.05, 63 leaves, ≥ 50 samples per leaf, 0.8 row/column subsampling.

Key design decisions. (1) PPR edges count ≥ 80 %-watched airings, because the label counts exactly those (log-seconds edges: best 0.3055 vs 0.3871, test users). (2) History counts are per-week rates (plus fixed-length last-4/2-week counts), because fold A has 10 history weeks and the final run 20; the final ranker is trained on fold A and fold B (14 weeks) candidate tables. (3) Candidates are restricted to shows that air in the target window — a show that is not broadcast cannot be watched.

4. Experiments

Method (best config of its grid)	MAP@5 fold A (all, n=4,643)	MAP@5 fold A (test, n=1,238)	fold B (test)	fit + predict	Kaggle public / private
Popularity (all history weeks)	0.2465	0.2688	0.2742	0.6 s	0.30258 / 0.29969
Popularity (last 4 weeks)	0.2747	0.3022	–	0.4 s	–
Repeat frequency (reference)	0.3201	0.3631	0.3413	1.8 s	–

Method (best config of its grid)	MAP@5 fold A (all, n=4,643)	MAP@5 fold A (test, n=1,238)	fold B (test)	fit + predict	Kaggle public / private
PPR α=0.7, edge=count80, restart=user	0.3460	0.3871	0.3576	9.9 s	0.41923 / 0.40848
ALS 64 factors, λ=0.1, conf=1+count80	0.1908	0.1923	0.2045	2.9 s	–
Rules sup≥20, conf≥0.2, max, self-rule	0.2739	0.2973	0.2904	1.8 s	–
Hybrid v1: LightGBM lambdarank, 4 candidate sources	0.3813	0.4214	–	57.8 s (CV) / 54.0 s (final)	–
+ channel-affinity candidates	0.3853	0.4258	–	63.0 s (CV)	–
+ within-user relative features	0.3862	0.4258	–	58.8 s (CV) / 61.6 s (final)	0.44985 / 0.44436
4-objective rank-average ensemble (final)	0.3903	**0.4319**	–	4 CV runs 49–63 s each / 136.6 s (final)	**0.45214 / 0.44826**

Table 2. Same fold, users, candidates and metric for every row (`results/attempts.csv`, 63 runs incl. 12 PPR, 6 ALS, 10 rules, 27 hybrid runs; `results/timings.csv`; CV times include the 22 s candidate build). Not chosen (test users): v1 binary 0.4172 / 3 seeds 0.4195; fold-B rows in CV training 0.4255; recency features with lambdarank alone 0.4232; Poisson / graded / binary alone 0.4213 / 0.4248 / 0.4219; 1,000 trees and 127 leaves 0.4166; channel top-100 candidates (recall 0.8662) 0.4282.

Comparison vs baseline. Every personalised method except ALS beats popularity; PPR adds +0.118 MAP@5 (test users) over popularity and +0.024 over the user's own repeat frequency, and the re-ranker ensemble a further +0.045. Of that, channel-affinity candidates give +0.004 (candidate recall 0.7909 → 0.8428) and rank-averaging four objectives +0.006, a gain that repeats on two other feature sets (0.4292 vs 0.4258; 0.4282 vs 0.4246); the within-user relative and recency features give nothing measurable. Fold B keeps the single-method order on both user sets (all users: PPR 0.3236 > repeat 0.3078 > rules 0.2667 > popularity 0.2479 > ALS 0.1954). Channel-affinity and PPR rank + score carry 34 % and 30 % of the split gain (`results/feature_importance_chan+rec+rel.csv`).

Two users (Table 3). User A stands for heavy viewers: among the 10 % heaviest test users it is the one on whom the methods disagree most (18 distinct programmes across the six Top-5 lists), so the differences between methods are visible. User B stands for the near-cold segment (2 airings watched ≥ 80 % in weeks 11–20), where history-based methods have almost nothing to work with. Both were picked by a fixed rule in `src/examples.py`, not by looking at scores. Same users and same fold-A protocol for every method; ✓ = in the fold-A truth; full tables with genres in `results/examples.md`.

User A (heavy), `5024631139882254261`. History weeks 11–20 (airings ≥ 80 %): Вспыш и чудо-машинки [Детям] 143; Свинка Пеппа [Детям] 140; Маша и Медведь [Детям] 66; Моланг [Детям] 48. Truth weeks 21–30: 12002955 Свинка Пеппа [Детям]; 12002896 Вспыш и чудо-машинки [Детям]; 12002805 Моланг [Детям]; 12002172 Гуппи и пузырики [Детям]; 12002806 Октонавты [Детям].

method	AP@5	1	2	3	4	5
popularity	0.000	2400480 ТСН. Неделя [Инфо]	240081 Факты. День [Инфо]	2400467 Викна-Новости [Инфо]	200344 Чистоnews [Развлечения]	2400508 Сегодня. Итоги с Олегом… [Инфо]
repeat	0.550	✓ 12002896 Вспыш и чудо-машинки [Детям]	✓ 12002955 Свинка Пеппа [Детям]	12002201 Маша и Медведь [Детям]	✓ 12002805 Моланг [Детям]	12001682 Губка Боб Квадратные Шт… [Детям]
ppr	0.400	✓ 12002955 Свинка Пеппа [Детям]	✓ 12002896 Вспыш и чудо-машинки [Детям]	12001682 Губка Боб Квадратные Шт… [Детям]	2400480 ТСН. Неделя [Инфо]	12002201 Маша и Медведь [Детям]
als	0.000	10002514 Кухня [Сериалы]	200469 Варьяты. Лучшее [Развлечения]	12001732 Пингвины из Мадагаскара [Детям]	12002456 Пчёлка Майя [Детям]	6700482 Секретные материалы [Другое]
rules	0.000	2400480 ТСН. Неделя [Инфо]	240081 Факты. День [Инфо]	2400467 Викна-Новости [Инфо]	200344 Чистоnews [Развлечения]	20088 Завтрак с 1+1. Информац… [Развлечения]
hybrid	0.520	✓ 12002955 Свинка Пеппа [Детям]	✓ 12002896 Вспыш и чудо-машинки [Детям]	12001682 Губка Боб Квадратные Шт… [Детям]	12002856 Элвин и бурундуки [Детям]	✓ 12002172 Гуппи и пузырики [Детям]

User B (near-cold), `11191705376024801445`. History weeks 11–20 (airings ≥ 80 %): Голос страны [Развлечения] 2; ТСН. Неделя [Инфо] 0; МастерШеф [Познавательное] 0; Ученик чародея [Фильмы] 0. Truth weeks 21–30: 10002514 Кухня [Сериалы]; 90039375 Бриллиантовый полицейск… [Фильмы]; 90066875 Чистильщик [Фильмы]; 90075419 Премьера. Х/ф Клин клин… [Фильмы]; 90076668 Два ствола [Фильмы].

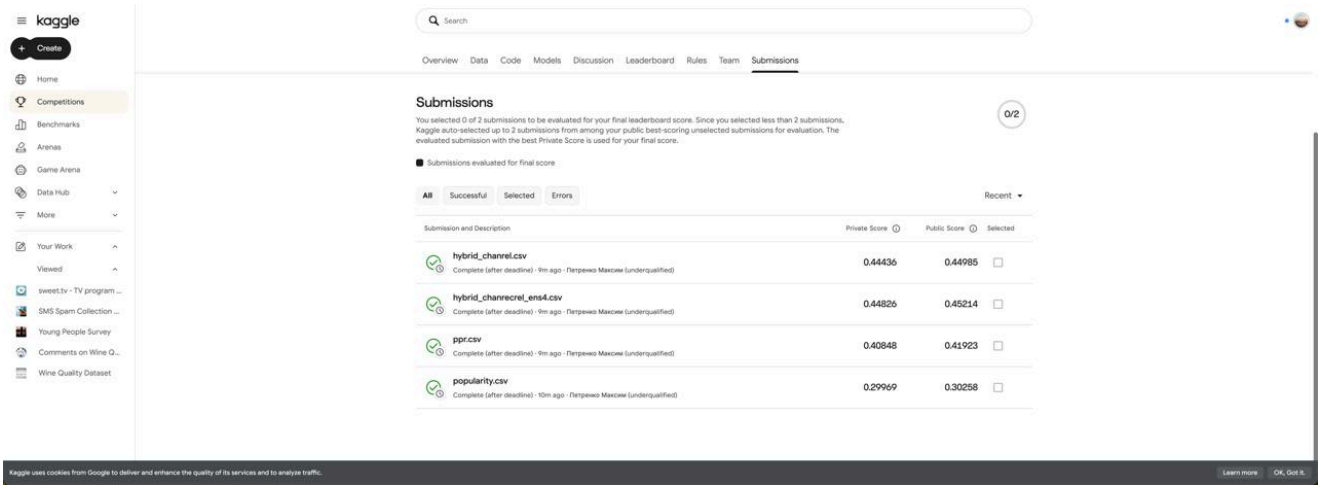
method	AP@5	1	2	3	4	5
popularity	0.000	2400480 ТСН. Неделя [Инфо]	240081 Факты. День [Инфо]	2400467 Викна-Новости [Инфо]	200344 Чистоnews [Развлечения]	2400508 Сегодня. Итоги с Олегом… [Инфо]
repeat	0.040	2400480 ТСН. Неделя [Инфо]	400361 МастерШеф [Познавательное]	90071909 Ученик чародея [Фильмы]	90078669 Терминатор: Генезис [Фильмы]	✓ 10002514 Кухня [Сериалы]
ppr	0.000	2400480 ТСН. Неделя [Инфо]	240081 Факты. День [Инфо]	12001682 Губка Боб Квадратные Шт… [Детям]	20088 Завтрак с 1+1. Информац… [Развлечения]	2400467 Викна-Новости [Инфо]
als	0.000	500359 Свобода слова [Развлечения]	700382 Рассмеши комика [Развлечения]	12002201 Маша и Медведь [Детям]	400426 Меняю жену [Развлечения]	200405 Мамахохотала-шоу. Лучшее [Развлечения]
rules	0.000	2400480 ТСН. Неделя [Инфо]	240081 Факты. День [Инфо]	200344 Чистоnews [Развлечения]	2400467 Викна-Новости [Инфо]	2400508 Сегодня. Итоги с Олегом… [Инфо]
hybrid	0.000	2400480 ТСН. Неделя [Инфо]	12001682 Губка Боб Квадратные Шт… [Детям]	12002955 Свинка Пеппа [Детям]	2400508 Сегодня. Итоги с Олегом… [Инфо]	12001734 Волшебные покровители [Детям]

Table 3. Top-5 per method (programme ID, title, [category]) for the two users, fold A.

User A's history is children's cartoons: repeat, PPR and the hybrid stay in [Детям] and hit 2–3 truth items, and the hybrid also finds 'Гуппи и пузырики' (✓), which no single method ranks in its Top-5. ALS leaves the user's category (series, entertainment, 'Секретные материалы') — what similar households watch, not what this one repeats; rules and popularity return the same news block [Инфо] that everybody gets (in rules all the user's own shows tie at the self-rule confidence 1 and popularity breaks the tie). User B watched only 'Голос страны' ≥ 80 %, which does not air in weeks 21–30, and later watched films [Фильмы] and 'Кухня' [Сериалы]; only repeat finds 'Кухня' (rank 5), the other methods fall back to news and cartoons — nothing in the history points to films.

Verification. (i) `tools/evaluate.py` self-test passes (`results/evaluate_selftest.txt`); (ii) PPR closed form = power iteration to 7.0e-10; (iii) one join (view 20:59:15–22:04:50 on channel 1495 overlaps three cartoon airings by 1,200, 1,500 and 1,190 s) and (iv) one AP@5 are laid out in `results/hand_check.md`; (v) every submission passes `tools/check_submission.py`; (vi) the four uploaded files keep their fold-A order on Kaggle's private split (0.44826 > 0.44436 > 0.40848 > 0.29969 vs 0.4319 > 0.4258 > 0.3871 > 0.2688, test users), each 0.016–0.031 higher on Kaggle (`results/kaggle_scores.csv`). Items (iii) and (iv) are the hand checks listed in §6.

Kaggle (Figure 1). Designated final submission, chosen by fold-A MAP@5 (`results/final_choice.md`): `submissions/hybrid_chanrecrel_ens4.csv`; private score 0.44826, public 0.45214 — the best of the four uploads.



5. Discussion

Strategy for sweet.tv. We would ship the hybrid: it is the best method on validation (MAP@5 0.4319 vs 0.3871 for PPR and 0.2688 for popularity, test users) and on Kaggle's private split (0.44826 vs 0.40848 and 0.29969), and its cost is negligible for a weekly batch (136.6 s to fit and score 1,258 users). It is, however, five candidate generators and four models. If simplicity matters more than the last points, PPR with count-weighted edges is the fallback: one hyper-parameter, 9.9 s for 4,643 users, and 91 % of the hybrid's private score. Table 3 shows why either beats popularity: user A's household watches cartoons, and the popularity list offers five news programmes.

How the methods differ and what users would see. The label counts *repeated* viewing, and the methods differ mainly in how much they repeat the user's own shows. Repeat frequency and rules use 98.8 % of the slots for shows the user already watched ≥ 80 %, PPR 93.8 %, the hybrid 89.2 % (`results/ux_stats.csv`) — a "your shows" row with high precision but little discovery. ALS reconstructs what similar households watch and leaves the user's taste (user A gets 'Кухня', 'Варьяты', 'Секретные материалы', AP@5 0), which costs accuracy (0.1923, below popularity). Popularity shows the same 5 programmes to everybody; the hybrid uses 112 distinct shows, PPR 190, repeat 438. All personalised lists are more repeat-heavy than real behaviour: only 54.4 % of a user's future top-5 had been watched ≥ 80 % before and 28.5 % were never touched (`results/truth_repeat_share.txt`). For the product this suggests a separate discovery slot rather than more novelty inside the top-5, since only 2.8 % of the hybrid's slots are new to the user.

Failure case. The first `python src/run.py final` wrote the single-method files and then produced no output for more than 10 minutes. A first stack sample pointed at LightGBM (a red herring); a second one showed a Python loop inside pandas `IndexEngine.__contains__`. Root cause: `np.array([u for u in FB.users_all if u in set(FB.hist.uid)])` rebuilt a set of all history rows for every fold-B user — quadratic work hidden in one line. Fix: `np.intersect1d(FB.users_all, FB.hist.uid.unique())`; the rerun finished in 58 s and the file passed the format checker (`results/final_log.txt`, `results/decisions.md`). A second, process-level failure: while drafting, the AI wrote fold-B numbers that are not in `results/attempts.csv` and a Kaggle-gap range that did not match the scores; both were caught by checking every number against its source file and corrected.

What worked. (i) Edge weights that mirror the label: counting ≥ 80 %-watched airings instead of log-seconds raised PPR from 0.3055 to 0.3871. (ii) Schedule-aware candidates: adding shows from the user's channels that air in the target window raised candidate recall from 0.7909 to 0.8428. (iii) Rank-averaging four objectives (binary, lambdarank, graded lambdarank, Poisson on the count): +0.006 on validation, repeated on two other feature sets, and +0.0039 on Kaggle private (0.44826 vs 0.44436).

What did not work or surprised us. ALS stayed below popularity in every configuration; a larger confidence scale collapsed it to 0.0321. Lift-scored rules fell to 0.0688 because lift promotes rare shows. A bigger LightGBM (0.4166), recency features alone (0.4232) and channel top-100 candidates (recall 0.8662, MAP@5 not higher) did not help. Kaggle scores were 0.016–0.031 above validation for all four uploads, with the order preserved.

Limitations. About 16 % of the truth items are outside the candidate set, so no ranker can reach them. Near-cold users are barely served: for user B every method except repeat scores 0. The label rule is reconstructed from the Data-tab text and its tie-break is unknown. Fold A has 10 history weeks while the final run has 20; we normalise counts per week but did not test whether this explains the Kaggle–validation gap. Only 53.1 % of watched seconds fall on labelled airtime, and the timezone check is only weakly conclusive. With 1,238 test users we treat differences of a few thousandths between configurations as noise.

Next improvement. A content-based candidate source for shows the user has never watched — same category/genre as the user's history and airing on the user's channels — aimed at the 28.5 % of future favourites that history-based methods cannot reach.

6. AI Usage Disclosure

- Tools: Claude Code 2.1.316 (model Claude Opus 5.5), plus Claude (chat) for preparing the task kit (CLAUDE.md, PROMPT.md, tools/, references.md).
- How AI was used: code generation (all of `src/`), debugging (Kaggle CLI 403, the quadratic `set()` hang in `run.py final`), data checks and report drafting (all numbers copied from `results/`); search was not used — references come from the pre-vetted `references.md`.
- Report: drafted by Claude Code from the files in `results/` (the instructor allows AI-drafted reports if the student hand-checks them); I checked it as listed below.
- What I personally verified: (1) AP@5 = 0.520 for user A recomputed by hand from `results/hand_check.md`; (2) the three join overlaps 1,200 / 1,500 / 1,190 s recomputed from the view record and airing times; (3) the ground-truth rule in §1 matches the competition Data tab; (4) every reference URL opens; (5) the Kaggle scores in Table 2 match my Submissions page, Figure 1; (6) re-ran `python src/run.py validate` and got the same fold-A numbers.
- What I trusted without verification: the join and feature code beyond the single worked example; the PPR closed form beyond the numerical check against power iteration; the internals of `implicit` and LightGBM; the tie-break of the truth rule; the explanation of the Kaggle–validation gap.
- Session log: `petrenko_maksim_midterm_session.json` (Claude Code transcript, unmodified), 2026-11-01, 13:22 – 17:17 (MSK).

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